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Batch:07

```
def rectangle_metrics(length, width):  
    """  
    Calculates area and perimeter of a rectangle.  
  
    Parameters:  
    length (float): Length of rectangle  
    width (float): Width of rectangle  
  
    Returns:  
    tuple: (area, perimeter)  
    """  
    area = length * width  
    perimeter = 2 * (length + width)  
    return area, perimeter  
  
rectangles = [(5, 10), (7, 12), (10, 15)]  
  
for length, width in rectangles:  
    area, perimeter = rectangle_metrics(length, width)  
    print("Area:", area)  
    print("Perimeter:", perimeter)  
    print()
```

Area: 50
Perimeter: 30

Area: 84
Perimeter: 38

Area: 150
Perimeter: 50

```
names = ["Alice", "Bob", "Charlie", "David"]  
search_names = ["Charlie", "Eve", "Bob"]  
  
name_set = set(names)  
  
for s in search_names:  
    if s in name_set:  
        print(f"{s} is in the list")  
    else:  
        print(f"{s} is not in the list")
```

Charlie is in the list
Eve is not in the list
Bob is in the list

```
def calculate_total(price):  
    """  
    Calculates total price including 18% tax.  
  
    Parameters:  
    price (float): Base price  
  
    Returns:  
    float: Total price including tax  
    """  
    TAX_RATE = 0.18  
    return price + (price * TAX_RATE)  
  
prices = [250, 500]  
  
for price in prices:  
    print("Total Price:", calculate_total(price))
```

Total Price: 295.0
Total Price: 590.0

PI = 3.14159
RADIUS = 7

```
area = PI * (RADIUS ** 2)
circumference = 2 * PI * RADIUS

print("Area of Circle:", area)
print("Circumference of Circle:", circumference)
```

```
Area of Circle: 153.93791
Circumference of Circle: 43.98226
```

```
base = 10
height = 20

# Area of triangle formula
area_of_triangle = base * height / 2

print("Area of Triangle:", area_of_triangle)
```

```
Area of Triangle: 100.0
```